

ABDULLAH AL MAMUN

AI & Computer Vision Researcher | Self-Supervised Learning | Vision Mamba | Precision Agriculture

Brisbane, QLD, Australia | +61 420 846 485 | mamun130203@gmail.com | a.mamun@griffith.edu.au | mam012@csiro.au

RESEARCH PROFILE

Doctoral researcher in artificial intelligence, computer vision, and self-supervised learning at Griffith University in collaboration with CSIRO Data61. Research focuses on data-efficient plant disease recognition, Vision Mamba/state-space models, contrastive learning, teacher-student distillation, and robust visual representation learning. Current publication record includes 54 peer-reviewed outputs (36 journal articles and 18 international conference papers/book chapters), with 822 Google Scholar citations, h-index 16, and i10-index 27 as reported in the current academic CV. First author of recent work in Pattern Recognition and Expert Systems with Applications, with additional research presented at the AAAI 2026 AgriAI Workshop. Experienced in multi-GPU deep-learning experimentation, research-led teaching, academic leadership, and cross-institutional collaboration.

RESEARCH SNAPSHOT

54	822	16	3
Peer-reviewed outputs	Google Scholar citations	h-index	Recent first-author AI papers (2026)

RESEARCH EXPERTISE

- Self-supervised learning, contrastive learning, knowledge distillation, teacher-student learning, and representation learning.
- Vision Mamba and state-space models for efficient long-range visual modelling and multi-scale feature learning.
- Plant disease detection and recognition, precision agriculture, domain-robust computer vision, and data-scarce learning.
- Deep learning for image segmentation and recognition, predictive modelling, sensing-oriented applications, and quantitative analysis.
- Reproducible experimental design, ablation studies, benchmarking, statistical evaluation, scientific writing, and peer-reviewed publication.

EDUCATION

Doctor of Philosophy (PhD), Information and Communication Technology | Griffith University **Oct 2023 - Present**
in association with CSIRO, Australia

- Thesis: Plant Disease Detection Using Self-Supervised Learning.
- Supervision: Dr Mohammad Awrangjeb, Dr David Ahmedt-Aristizabal, Dr Miaohua Zhang; CSIRO collaborator/supervisor Dr Zeeshan Hayder.
- Research emphasis: self-supervised representation learning, Vision Mamba/state-space modelling, efficient plant disease recognition, and robust evaluation across multiple datasets.

Master of Engineering Science (by Research) | Multimedia University, Malaysia **Sep 2019 - Jun 2021**

- Thesis: Lane Markings Detection Using Encode-Decode Instance Segmentation Network.
- Supervisor: Dr Em Poh Ping.

Bachelor of Science in Electrical and Electronic Engineering | Pabna University of Science and Technology, Bangladesh **Jan 2013 - Oct 2018**

- CGPA: 3.87/4.00; merit position: 1/46.
- Thesis: Small Intestine Bleeding Detection Using Wireless Capsule Endoscopy.

RESEARCH EXPERIENCE AND CONTRIBUTIONS

Doctoral Researcher - AI, Computer Vision & Precision Agriculture | Griffith

2023 - Present

University / CSIRO Data61, Australia

- Developed self-supervised learning frameworks for plant disease detection using Vision Mamba and state-space sequence modelling.
- Designed contrastive, prototype-driven, multi-scale and hierarchical teacher-student objectives to improve disease-aware feature learning from limited labels.
- Conducted controlled benchmarking across public plant disease datasets, including ablation studies, robustness analysis, feature-space visualisation, and comparative evaluation against CNN-, Transformer-, SSL-, and Mamba-based baselines.
- Implemented and trained models in PyTorch using CUDA-enabled multi-GPU workflows and distributed data-parallel training; maintained reproducible research pipelines for large experimental matrices.
- Collaborated with Griffith and CSIRO researchers across model design, experimental validation, manuscript preparation, conference presentation, and publication.

Researcher - Deep Learning for Intelligent Transportation | Multimedia University,

2019 - 2021

Malaysia

- Developed an encode-decode instance-segmentation framework for lane marking detection under varying road and environmental conditions.
- Evaluated performance across TuSimple, Caltech, and locally collected data, supporting a peer-reviewed Heliyon publication.

SELECTED PUBLICATIONS

1. A. A. Mamun, M. Zhang, D. Ahmedt-Aristizabal, Z. Hayder, and M. Awrangjeb, "ConMamba: Contrastive Vision Mamba for Plant Disease Detection," Pattern Recognition, vol. 176, 113177, 2026. DOI: 10.1016/j.patcog.2026.113177. **First author**
2. A. A. Mamun, M. Zhang, D. Ahmedt-Aristizabal, Z. Hayder, and M. Awrangjeb, "PSMamba: Progressive Self-supervised Vision Mamba for Plant Disease Recognition," Expert Systems with Applications, article 133775, available online 30 July 2026 (Volume 333, Part B, 2027). DOI: 10.1016/j.eswa.2026.133775. **First author**
3. A. A. Mamun, M. Zhang, D. Ahmedt-Aristizabal, Z. Hayder, and M. Awrangjeb, "StateSpace-SSL: Linear-Time Self-supervised Learning for Plant Disease Detection," AgriAI Workshop, AAAI 2026; arXiv:2512.09492. **First author; accepted & presented**
4. A. Al Mamun, D. Ahmedt-Aristizabal, M. Zhang, M. I. Hossen, Z. Hayder, and M. Awrangjeb, "Plant Disease Detection Using Self-Supervised Learning: A Systematic Review," IEEE Access, vol. 12, pp. 171926-171943, 2024. DOI: 10.1109/ACCESS.2024.3475819. **First author**
5. M. I. Hossen, M. Awrangjeb, S. Pan, and A. Al Mamun, "Transfer Learning in Agriculture: A Review," Artificial Intelligence Review, vol. 58, article 97, 2025. DOI: 10.1007/s10462-024-11081-x.
6. A. Al Mamun, E. P. Ping, M. J. Hossen, B. Jahan, and A. Tahabilder, "A Deep Learning Approach for Lane Marking Detection Applying Encode-Decode Instant Segmentation Network," Heliyon, vol. 9, no. 3, e14212, 2023. DOI: 10.1016/j.heliyon.2023.e14212. **First author**
7. M. F. Ali, M. R. I. Sheikh, A. A. Mamun, and M. J. Hossen, "Techno-economic, Predictive Modeling, and Demand Response Analysis of a Renewable Energy-Based Microgrid for Residential Applications," IEEE Access, vol. 13, pp. 53748-53771, 2025. DOI: 10.1109/ACCESS.2025.3552056.
8. M. R. Ali, M. F. Ali, D. Biswas, A. A. Mamun, and M. J. Hossen, "Sustainable Energy Solutions for Rural Bangladesh: An Optimized Hybrid Microgrid Model," Frontiers in Energy Research, vol. 13, 2025. DOI: 10.3389/fenrg.2025.1652536.

Publication profile: 54 peer-reviewed outputs, including 36 journal articles and 18 international conference papers/book chapters. Full publication list available via Google Scholar or on request.

CURRENT AND RECENT RESEARCH DIRECTIONS

- Efficient self-supervised learning using Vision Mamba/state-space models for high-resolution and data-scarce visual recognition.
- Multi-scale lesion representation, hierarchical distillation, and local-global alignment for plant disease recognition.
- Robust and generalisable AI for precision agriculture, including domain shift, few-label settings, and real-world deployment constraints.
- Ongoing manuscript development on advanced self-supervised representation learning beyond current ConMamba, PSMamba, and StateSpace-SSL frameworks.

ACADEMIC APPOINTMENTS AND TEACHING

- Sessional Academic / Lecturer - ICT705 Database Systems Analysis and Design** | **Sep 2026 - Present**
Australian Institute of Business Intelligence (AIBI), Brisbane, Australia
- Deliver postgraduate teaching in database systems and design, including data modelling, relational design, SQL, transaction concepts, and contemporary database technologies.
- HDR Teaching Fellow (Associate Lecturer)** | *School of Information and Communication Technology, Griffith University, Australia* **Mar 2026 - Present**
- Support research-led ICT teaching, tutorials, assessment, student consultation, and learning support.
- Casual Academic** | *School of Information and Communication Technology, Griffith University, Australia* **Mar-Jun 2024; Jul 2025-Feb 2026**
- Contributed to tertiary teaching, assessment activities, student support, and feedback in ICT courses.
- Assistant Professor and Chairman** | *Department of Electrical and Electronic Engineering, Feni University, Bangladesh* **Jan 2023 - Jan 2024**
- Led teaching, academic coordination, staff communication, student support, curriculum delivery, and departmental administration.
- Lecturer and Chairman (In-charge from Feb 2021)** | *Department of Electrical and Electronic Engineering, Feni University, Bangladesh* **Jan 2019 - Dec 2022**
- Delivered undergraduate engineering teaching and research supervision while coordinating departmental academic responsibilities.
- Chief Instructor** | *Electrical Department, Mirpur Institute of Science and Technology, Bangladesh* **May 2018 - Dec 2018**

TECHNICAL AND RESEARCH SKILLS

Programming & ML	Python, PyTorch, C, C++, MATLAB; deep learning model development and quantitative analysis.
AI / Computer Vision	Self-supervised learning, contrastive learning, Vision Mamba/state-space models, CNNs, Transformers, segmentation, classification, transfer learning.
Compute	CUDA-enabled GPU training, multi-GPU/distributed data-parallel workflows, Linux research environments, experiment tracking and reproducibility.
Research Methods	Benchmark design, ablation studies, statistical comparison, literature review, systematic review, scientific visualisation, reproducible pipelines.
Academic Tools	LaTeX, EndNote, Mendeley, MS Word, PowerPoint, Excel, Visio, draw.io, Lucidchart.
Teaching	AI/ML and computing concepts, database systems, engineering education, assessment, feedback, student mentoring, academic coordination.

SCHOLARSHIPS, GRANTS, AWARDS AND SERVICE

- CSIRO Top-Up Scholarship, Australia (2023-2026).
- Griffith University Postgraduate Research Scholarship (2023-2026).
- Griffith University International Postgraduate Research Scholarship (2023-2026).
- Fi Sabilillah Research Development Grant Scheme, Malaysia (2021).
- Fundamental Research Grant Scheme, Malaysia (2019-2020).
- President, Griffith University Bangladesh Association (2024-2025).
- Founder, Research Association, Pabna University of Science and Technology.
- Member, IEEE.